

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 - Case Study Analysis

Course Code:	CSA10/CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 - Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	Tharun	Reg. No.:	1911260024
Date:	4/10/26	Duration:	60 minutes

Code of Conduct

I Tharun - 1911260024 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Tharun*

Annexure A - Case Study Analysis

Instructions to Students:

Each student will be allotted one problem statement. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

Development of a Software Program/Application for Generation of Test Reports for Non-Automatic Weighing Instruments (NAWI) as per OIML Recommendation R-76

1. Understand the Problem Statement

1.1 Explain the problem in your own words.

Non-Automatic Weighing Instruments (NAWIs)—such as electronic retail balances, precision laboratory scales, industrial platform scales, and heavy-capacity weighbridges—require operator intervention during the weighing process and are widely used in trade, commerce, healthcare, agriculture, and industry where accurate measurement is essential for fair transactions and consumer protection. Under the Legal Metrology Act, 2009 and the Legal Metrology (General) Rules, 2011, every NAWI model used for commercial transactions and protection must conform to prescribed standards and obtain formal Model Approval (Type Evaluation) from designated testing laboratories, such as Regional Reference Standard Laboratories (RRSLs), before undergoing verification and stamping.

These type evaluation procedures are governed internationally by OIML Recommendation R-76 (Non-Automatic Weighing Instruments - Part 1: Metrological and technical requirements - Tests, and Part 2: Test report format). At present, these test reports are largely prepared manually by recording observations on paper datasheets or isolated spreadsheets and transcribing the results into static document templates. Engineers must manually compute changeover points, unrounded indication errors, corrected errors, and class-specific Maximum Permissible Errors (MPEs) across hundreds of test load points. This manual process is time-consuming, prone to calculation and transcription errors, and lacks uniformity across laboratories.

The proposed system is intended to support authorized Legal Metrology testing laboratories by automating test data recording, metrological error calculations, rule-based compliance evaluation, and the generation of standardized digital test reports strictly adhering to OIML R-76-2.

1.2 Identify the Objectives of the proposed system

1. Capture comprehensive instrument details, manufacturer information, model specifications, and metrological parameters (Class I, II, III, or IIII, Max, Min, e, and d).
2. Record laboratory testing infrastructure details, reference standard weight traceability, and prevailing environmental conditions (ambient temperature, pressure, and humidity).
3. Provide structured digital data entry forms for all prescribed OIML R-76 test procedures across single-interval, multi-interval, and multiple-range instruments.
4. Automatically calculate changeover points, unrounded indication errors (E), zero-corrected errors (Ec), and temperature sensitivity coefficients.
5. Automatically determine dynamic Maximum Permissible Error (MPE) limits based on accuracy class and load range, and evaluate pass/fail compliance.
6. Perform real-time input validation checks to detect missing entries, out-of-range test loads, or invalid environmental shifts during testing.
7. Generate standardized, print-ready digital test reports in strict accordance with OIML R-76-2 in both PDF and editable MS Word (DOCX) formats.
8. Maintain a centralized, searchable digital repository of completed test reports, instrument-wise test histories, and attached photographs.
9. Provide secure user access with role-based permissions and a dashboard for monitoring ongoing testing activities and approval status.
10. Support configurable rule tables so future revisions to OIML recommendations can be incorporated without rewriting core software code.

1.3 State the Scope of the problem

- Instrument specification and manufacturer application intake management.
- Laboratory environmental condition and reference standard equipment logging.
- Digital observation capture for metrological performance tests (weighing performance, hysteresis, eccentricity, repeatability, discrimination, creep, zero-return, and tare).
- Digital observation capture for influence factor and disturbance tests (tilting, warm-up time, static temperatures, damp heat, mains voltage variation, electrical bursts, electrostatic discharge, and electromagnetic immunity).
- Automated calculation of permissible errors, indication errors, and pass/fail status.
- Attachment of instrument photographs, load-receptor eccentricity sketches, and technical checklists.
- Standardized OIML R-76-2 test report generation and export in PDF and DOCX formats.
- Instrument-wise test history repository, multi-parameter search, and analytical dashboard.
- Role-based access control and digital signature support for authorized Legal Metrology laboratories.

2. Identify Key Stakeholders and Challenges

2.1 Identify all the stakeholders involved in the system

The main stakeholders involved in the OIML R-76 NAWI Test Report Generation System are:

- Metrological Testing Engineers (RRSLs)
- Laboratory Directors / Approving Officers
- Ministry of Consumer Affairs (Legal Metrology Division)
- NAWI Manufacturers & Importers
- Quality Assurance / NABL Auditors (ISO/IEC 17025)
- System Administrators
- Software Developers & Data Engineers
- Commercial Users & Consumers

2.2 Explain the Roles and Responsibilities of Each Stakeholder

Stakeholder	Roles and Responsibilities	Major Challenges
Metrological Testing Engineers (RRSLs)	Execute physical OIML R-76 tests, record load observations, and prepare draft test reports.	Tedious manual error calculations across hundreds of load points and high risk of formula mistakes.
Laboratory Directors / Approving Officers	Review completed test data, verify metrological compliance, and sign official reports.	Difficulty verifying thousands of spreadsheet cells manually and delayed review cycles.

Stakeholder	Roles and Responsibilities	Major Challenges
Ministry of Consumer Affairs (Legal Metrology)	Formulate policy, issue Model Approval Certificates based on lab reports, and oversee standards.	Lack of uniform reporting across regional laboratories and difficulty tracking turnaround times.
NAWI Manufacturers & Importers	Submit prototype weighing instruments for type evaluation and await certification.	Prolonged waiting periods for report generation and lack of visibility into testing progress.
Quality Assurance / NABL Auditors	Audit laboratory processes, calculation accuracy, equipment traceability, and record integrity.	Incomplete audit trails in manual spreadsheets and difficulty verifying raw data immutability.
System Administrators	Deploy, maintain, back up, and manage user access and security of the laboratory platform.	Ensuring high availability, role security, and smooth updates when OIML standards evolve.
Software Developers & Data Engineers	Design, implement, and validate the metrological rule engine, database, and report templates.	Translating dense OIML R-76 mathematical clauses and conditional checklists into software logic.
Commercial Users & Consumers	Rely on certified weighing instruments for fair trade, retail billing, and industrial weighing.	Risk of inaccurate transactions if non-compliant scales pass due to human reporting errors.

2.3 Major Challenges and issues faced by the stakeholders

1. Complex Metrological Error Formulas and Changeover Calculations: Digital indications require the changeover point method using supplementary weights ($dL = 0.1e$), where unrounded error $E = I + 0.5e - dL - L$ and corrected error $E_c = E - E_0$ must be calculated across hundreds of load points.
2. Multi-Interval and Multiple-Range Instrument Complexity: Dynamically shifting Maximum Permissible Error limits ($\pm 0.5e_i$, $\pm 1.0e_i$, $\pm 1.5e_i$) across partial weighing ranges (e_1 , e_2 , e_3) and four accuracy classes (Class I, II, III, IIII) is error-prone manually.
3. Strict Adherence to Standardized OIML R-76-2 Report Formatting: Maintaining the rigid tabular layout, standardized checkboxes (Passed, Failed, Not Applicable), and eccentricity receptor sketches manually in word processors often breaks table structures.
4. Environmental Condition Coupling and Zero-Drift Constraints: Verifying that ambient temperature shifts stay within bounds and zero-indication drift per 5 C (or 1 C for Class I) remains $\leq e$ requires time-correlated logging.
5. Data Integrity, Traceability, and Audit Compliance: ISO/IEC 17025 accredited laboratories require tamper-evident audit logs and standard weight calibration traceability, which isolated spreadsheets lack.
6. High Volume of Conditional Checklists: Determining which OIML R-76 clauses apply to battery-operated, mobile, or multi-tare instruments requires structured rule guidance.
7. Security of Proprietary Designs: Protecting confidential manufacturer schematics and load-cell data from unauthorized access.

3. Analyze the Current Approach

3.1 Describe the existing system or current process.

Currently, NAWI type evaluation and test report preparation at Legal Metrology laboratories are performed manually using paper observation sheets, isolated Microsoft Excel spreadsheets, and static Word templates. A testing engineer physically loads standard weights onto the instrument, records indications and supplementary changeover weights (dL), computes errors in spreadsheets, and manually copies the results and pass/fail ticks into a Word document formatted like OIML R-76-2.

3.2 Identify its strengths and limitations.

Strengths:

- Human metrologists directly observe physical instrument settling and display stability during testing.
- Laboratories use existing office software (Excel and Word) without custom software licensing fees.
- Engineers have flexibility to write unstructured notes for unusual instrument configurations.
- Paper sheets and local files work inside RF-shielded EMC chambers where wireless signals are blocked.

Limitations:

- Time-consuming manual transcription and report compilation taking several days per instrument.
- High risk of arithmetic and formula overwrite errors when computing E , E_c , and thermal zero-drift.

- Lack of real-time input validation during testing, leading to costly re-testing if a load point is missed.
- Inconsistent report layouts and rounding conventions across different regional laboratories.
- Test reports are scattered across individual desktop computers rather than a centralized searchable repository.
- Absence of immutable audit trails makes ISO/IEC 17025 accreditation audits difficult.

3.3 Analyze the problems or gaps that need to be addressed.

The major gap is the absence of an integrated, metrologically validated software system that combines guided OIML R-76 observation capture, automated changeover error and MPE calculation, immutable audit logging, and one-click OIML R-76-2 report export. Specifically, the following problems and gaps need to be addressed:

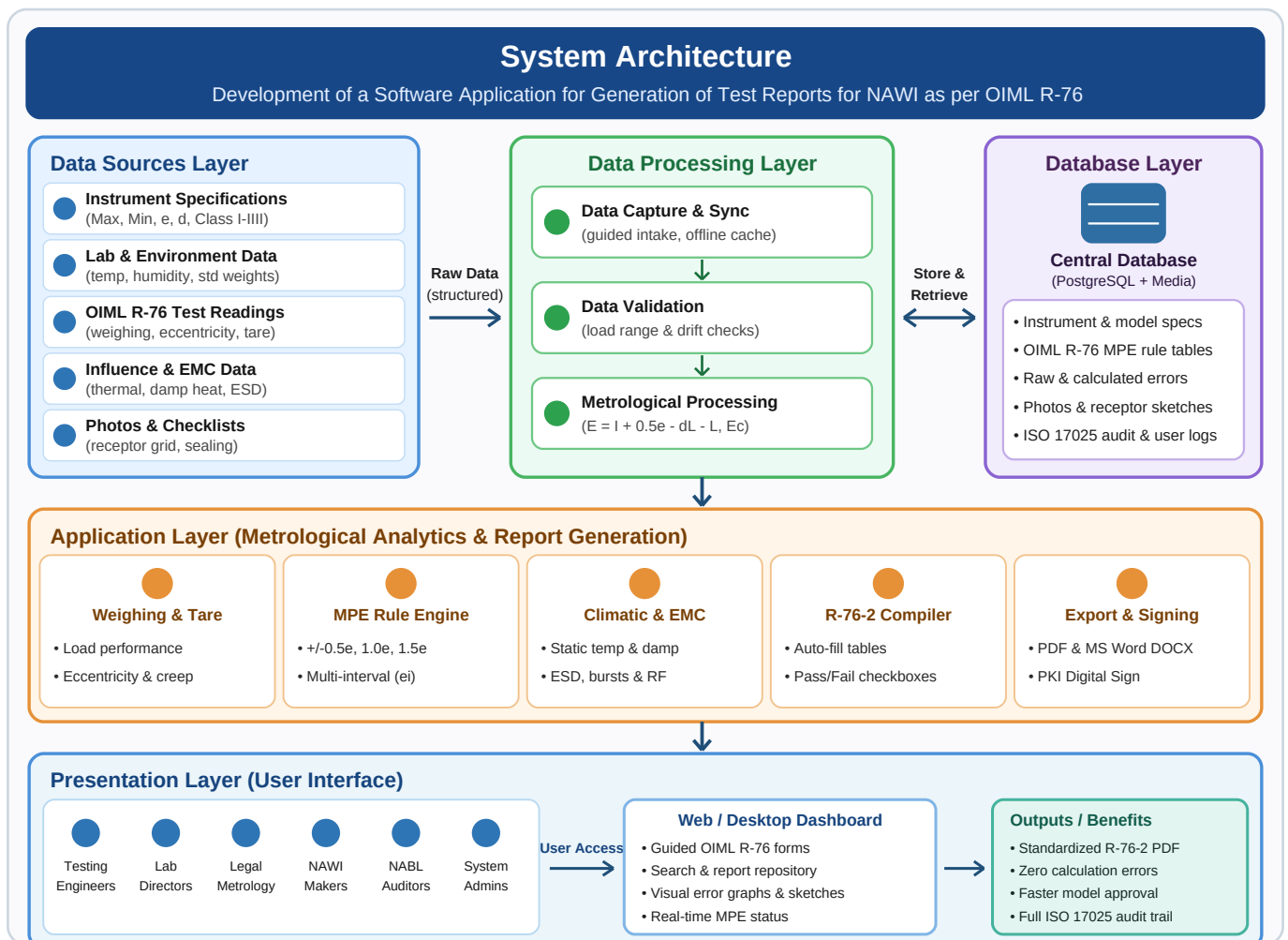
- Manual Error Calculation Mistakes - Formula overwrites in spreadsheets lead to inaccurate E and Ec values.
- Delayed Model Approval Turnaround - Re-typing hundreds of load points into Word tables causes severe bottlenecks.
- No Real-Time MPE Boundary Alerting - Engineers are not alerted immediately when an observation exceeds permissible limits during testing.
- Fragmented Historical Records - Lack of a centralized database prevents quick retrieval of past model approvals and test histories.

4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Design a software-based solution for the given problem.

To overcome the limitations of manual reporting, we propose the OIML R-76 Automated NAWI Test Report & Compliance Management System (NAWI-TRMS). The system calculates compliance and generates reports using:

Instrument Specs + Environmental Logs + Load Observations (I, L, dL) -> Error Calculation (E, Ec) + MPE Check -> Standardized OIML R-76-2 Report



The results can then be displayed through interactive observation tables, error curves, load-receptor eccentricity diagrams, and standardized OIML R-76-2 test reports.

4.2 Describe the major functional modules of the proposed system.

1. Instrument & Application Metadata Module - Captures manufacturer details, model specifications, accuracy class (Class I, II, III, IIII), Max, Min, e, d, and multi-interval configurations, automatically verifying $n = \text{Max} / e$.
2. Laboratory & Environmental Condition Module - Logs standard reference weights, calibration certificate traceability, and ambient temperature, pressure, and humidity before and after testing.
3. OIML R-76 Guided Observation Capture Module - Provides structured digital entry forms for Weighing Performance, Tare, Eccentricity, Repeatability, Discrimination, Creep, and Zero-Return tests.
4. Automated Metrological Calculation Engine - Computes changeover points, unrounded errors ($E = I + 0.5e - dL - L$), corrected errors ($E_c = E - E_0$), hysteresis, and thermal drift using exact decimal math.
5. MPE Rule Validation & Compliance Engine - Dynamically resolves Maximum Permissible Error limits ($\pm 0.5e$, $\pm 1.0e$, $\pm 1.5e$) and marks each load point and test section as Passed, Failed, or Not Applicable.
6. Influence Factor & Disturbance Module - Evaluates static temperatures, damp heat, voltage variations, bursts, ESD, and electromagnetic immunity tests for significant faults ($> e$).
7. Checklist, Sketch & Photo Module - Captures OIML R-76-2 technical checklists, interactive load-receptor eccentricity sketches, and instrument photographs.
8. Standardized OIML R-76-2 Report & Repository Module - Auto-populates official OIML R-76-2 templates, exports to PDF and MS Word (DOCX), supports PKI digital signatures, and maintains a searchable report repository.

4.3 Identify the functional and non-functional requirements.

Functional Requirements:

- User should be able to enter manufacturer details, instrument specifications, and metrological parameters.
- System should record laboratory environmental conditions and reference standard weight details.
- System should provide digital observation forms for all prescribed OIML R-76 tests.
- System should automatically calculate unrounded errors (E), corrected errors (Ec), and permissible errors (MPE).
- System should automatically determine Pass / Fail / Not Applicable compliance status.
- System should allow attachment of instrument photos, receptor sketches, and checklists.
- System should generate and export standardized OIML R-76-2 reports in PDF and editable DOCX formats.
- Administrator and directors should be able to review, digitally sign, and search historical reports via a dashboard.

Non-Functional Requirements:

- Usability: Clean, keyboard-efficient tabular interface with automatic load-point pre-filling.
- Performance: Real-time error calculation in under 100 ms and full PDF/DOCX report export within 5 seconds.
- Scalability: Support growing volumes of test records, photos, and multi-laboratory deployments across RRSLS.
- Security: Role-Based Access Control (RBAC), encrypted data storage, and PKI digital signatures.
- Reliability: Offline data entry inside RF-shielded EMC chambers with automatic synchronization.
- Maintainability: Configurable database rule tables allowing effortless updates when OIML R-76 is revised.
- Auditability: Immutable ISO/IEC 17025 activity logs recording every entry, modification, user ID, and timestamp.

4.4 Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

- Modularity -> The system is partitioned into independent modules (data intake, math engine, MPE validator, checklist manager, and PDF/DOCX exporter) for isolated testing and updates.
- Scalability -> Stateless backend services and normalized PostgreSQL + object storage allow expansion from a single laboratory to all regional laboratories nationwide.
- Maintainability -> Externalized OIML R-76 tolerance tables and report templates can be updated without modifying core application code.
- Reliability -> Strict input validation guards, local offline caching for shielded EMC rooms, and automated backups ensure zero data loss.
- Usability -> Guided workflows, automated changeover math, and visual out-of-tolerance alerts minimize engineer fatigue.
- Security -> Role-based permissions, encrypted storage, and cryptographic locking of signed reports prevent unauthorized tampering.

5. Justify the Chosen Methodology / Framework

5.1 Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

Selected SDLC Model: Agile (Scrum Framework)

The Agile software development methodology using the Scrum framework is selected for this project because the system involves multiple specialized OIML R-76 metrological test modules, complex multi-interval error calculations, and strict report layouts that benefit from iterative development and continuous validation by Legal Metrology domain experts across six 2-week Sprints.

5.2 Justify why the selected methodology/framework is suitable for the proposed system.

- Development is structured into 6 manageable Sprints (Sprint 1: Core & Specs; Sprint 2: Calculation Engine & Static Tests; Sprint 3: Climatic Tests; Sprint 4: EMC Disturbance & Offline Sync; Sprint 5: OIML R-76-2 PDF/DOCX Exporter; Sprint 6: ISO 17025 Audit & UAT).
- Individual OIML R-76 test modules can be developed, unit-tested against reference datasets, and verified separately.
- Continuous feedback can be collected from RRSL metrological testing engineers after each sprint demonstration.
- Complex multi-interval and subtractive tare edge cases can be refined early before final deployment.
- Automated regression testing (CI/CD) ensures 100% mathematical accuracy across all metrological formulas.

5.3 Explain how it supports successful project development and maintenance.

Requirement & OIML Clause Analysis -> Sprint Module Planning -> Rule Engine & UI Design -> Incremental Sprint Development -> Metrological Reference Testing -> Engineer Feedback -> Refinement -> Deployment & Maintenance

Support for Maintenance: Agile directly supports long-term maintenance because whenever OIML Recommendation R-76 or national Legal Metrology rules are revised, updated MPE tables and checklist clauses can be incorporated, regression-tested, and deployed through short maintenance sprints without disrupting ongoing laboratory operations.

6. Discuss Expected Outcomes and Limitations

6.1 Describe the expected benefits of the proposed solution.

The proposed system is expected to:

1. Drastically reduce the turnaround time required to compile and issue OIML R-76 type evaluation test reports.
2. Eliminate human arithmetic and transcription errors in calculating unrounded errors (E), corrected errors (Ec), and MPE limits.
3. Ensure 100% national uniformity in test report formatting in strict compliance with OIML R-76-2.
4. Provide real-time input validation during laboratory testing to immediately flag out-of-tolerance readings.
5. Establish a centralized, searchable digital repository of instrument-wise test histories and approved model certificates.
6. Simplify the evaluation of complex multi-interval and multiple-range weighing instruments.
7. Improve laboratory audit readiness through immutable ISO/IEC 17025 activity logs and calibration traceability.
8. Provide Laboratory Directors and Ministry officials with an analytical dashboard to monitor testing throughput.
9. Support instant export of finalized reports in both tamper-evident PDF and editable MS Word (DOCX) formats.

6.2 Explain how the solution addresses the identified challenges.

Identified Challenge	Proposed Software Solution
Complex changeover & error math (E, Ec)	Automated Metrological Calculation Engine using exact decimal arithmetic
Multi-interval & multi-range MPE shifts	Dynamic MPE Rule Validation Engine resolving partial ranges (e1, e2, e3) automatically
Broken tables in manual Word reports	Automated OIML R-76-2 Report Compiler exporting print-ready PDF and editable DOCX
Unnoticed ambient temperature drift	Environmental Logging Module with automated thermal zero-drift validation
Scattered files & slow retrieval	Centralized PostgreSQL Repository with multi-parameter search and dashboard
Lack of ISO/IEC 17025 audit trails	Immutable timestamped audit logger and PKI cryptographic digital signatures
No network inside RF-shielded EMC rooms	Hybrid offline local data caching with automatic server synchronization

6.3 Discuss potential risks, implementation challenges, and limitations.

Potential Risks & Implementation Challenges:

- Initial training overhead when transitioning veteran laboratory staff from manual spreadsheets to the new platform.
- Migrating legacy paper and Word test reports into the structured relational database requires initial effort.
- RF-shielded EMC chambers block live Wi-Fi, requiring reliable offline cache synchronization after testing.
- Centralizing proprietary manufacturer schematics requires strong cybersecurity and backup governance.

Limitations:

- For non-interfaced scales, accuracy still depends on the engineer correctly reading and typing the display indication (I) and supplementary weight (dL).
- The software reduces calculation and reporting time to seconds, but cannot shorten mandatory physical test durations (such as the 4-hour creep test or multi-day damp heat cycles).
- Unconventional prototype instruments with novel auxiliary devices may still require manual descriptive remarks.
- The system verifies calibration dates in the database but cannot physically detect if an uncalibrated weight is placed on the pan.

6.4 Suggest possible future enhancements.

- Direct hardware interfacing (RS-232, USB, Modbus) to automatically capture weight indications from NAWIs and telemetry from climatic chambers.
- Computer Vision / Optical Character Recognition (OCR) camera module to read seven-segment displays automatically during RF immunity tests.
- AI/ML anomaly detection to identify subtle load-cell creep patterns and thermal hysteresis trends across historical datasets.
- Direct API integration with national Legal Metrology portals (e-Maap and National Single Window System) for online certificate issuance.
- Blockchain-anchored QR codes on exported PDF reports for instant field authenticity verification.
- Automated calculation of expanded Uncertainty of Measurement (U) as per EURAMET cg-18 and ISO/IEC Guide 98-3 (GUM).
- Voice-assisted hands-free data entry allowing engineers lifting heavy weights to dictate observations directly.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25